

Data Driven Dynamic Decoupling Control for MIMO Precision Mechatronic Systems

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Abstract: Decoupling control is extensively employed in precision mechatronic systems, particularly within the semiconductor field. In this context, the inputs and outputs are effectively decoupled through the utilization of a static decoupling matrix. However, the prevalent challenge arises from cross-talk induced by errors inherent in the static decoupling process. This challenge poses a significant obstacle in meeting the stringent demands of high-speed and high-precision motion, stemming from inaccuracies in mechanical parameters and variations in motor characteristics, particularly under conditions of high dynamics. To enhance the performance of precision mechatronic systems, it is recommended to replace the decoupling matrix with a dynamic decoupling controller. In this paper, a data driven dynamic decoupling control method is developed to improve the decoupling accuracy by calibrating the mechanical parameters and compensating the differences of motors. The effectiveness and practicality of the proposed method are illustrated through numerical simulations.

Key Words: Data-based, Decoupling control, Motion control, Precision mechatronic systems.

1 Introduction

Precise mechatronic systems find widespread application in manufacturing machines and scientific instruments, as evidenced by their incorporation into various technologies such as industrial flatbed printers, CT scanners, atomic force microscopes, and more[1]. A state-of-the-art example of such a mechatronic system is the wafer scanner[2], which is used in the lithographic production of integrated circuits to achieve subnanometer positioning accuracy with high speed and acceleration. Precision mechatronic systems are characterized by multiple motion degrees-of-freedom (DOFs), ultra-precision and high dynamic motion, and delicate structural design. To fulfill the demands of multiple degrees-of-freedom motion, parallel design schemes, notably the H-bridge structure, are frequently employed. High-performance motion requires the utilization of advanced actuators, including linear motors, voice coil motors and piezoelectric ceramic motors. Simultaneously, a decoupling control strategy is implemented to diagonalize the relationship between reference input and system output[3].

Considering the mechatronic system from the perspective where the motor drive signal serves as the input and the displacement of the system performance point is chosen as the output, it is observed that the displacement of each degree of freedom can be influenced by the force exerted by any motor. Traditionally, it has been assumed that the mechanical system's stiffness and the motor's reproducibility are sufficiently high, implying the satisfaction of the rigid body hypothesis. This assumption further implies strict consistency between the motor instruction and the actual force exerted by the motor[4]. Consequently, a static decoupling matrix can be obtained to decouple the system completely. Nevertheless, the static decoupling matrix hinges on some mechanical installation parameters, such as the position of the center of mass (CoM), the position of the motor installation, etc. Simultaneously, variations in the manufacturing process introduce errors, leading to inconsistencies in the dynamic characteristics of different motors. Additionally, the dynamic characteristics of the same motor undergo changes

during high dynamic movement. Inaccuracies in mechanical parameters and disparities in dynamic characteristics among motors can diminish the decoupling accuracy of the system utilizing a static decoupling matrix. Consequently, the interactions of the system is aggravated, and the movement of one DOF of the system will inadvertently affect other DOFs which are remain positioning, and the errors occur in other DOFs refer to cross-talk. As accuracy requirements continue to advance, cross-talk has emerged as the primary impediment to achieving the desired performance.

To mitigate cross-talk, two methods can be employed. The first involves enhancing the system's ability to suppress cross-talk through the utilization of an advanced controller. Treating cross-talk as an external disturbance to the current degree of freedom of motion, various control methods with the ability to suppress disturbances can be applied. Examples include model predictive control, robust control, and feedforward control methods. Some specific methods have also been developed, and the most important of these is the equivalent transfer function (ETF) method. In [5], the interactions among DOFs are regard as parasitic dynamics and the equivalent transfer function (ETF) of each DOF is determined with which the cross-talk can be suppressed by special feedback controller. Nevertheless, the achievable bandwidth is limited by the mechanical structure. To avoid this limitation, a hybrid integrator-gain method[6] is developed which can achieve better low-frequency disturbance suppression. Nevertheless, the complexity of the controller design increases with the addition of regulatory parameters, leading to only a limited reduction in cross-talk. Consequently, the most direct and effective approach to address this issue is to enhance the precision of the decoupling matrix.

As discussed earlier, achieving complete decoupling of the system throughout the entire motion process necessitates not only enhancing the accuracy of mechanical parameters but also compensating for motor characteristics. This ensures consistency in characteristics between different motors and the same motor under various operating states[7]. The conventional method to improve the accuracy of the decou-

pling matrix used in industrial is to calibrate the mechanism parameters using professional measuring instruments. However, the actual measurement process faces two challenges. Firstly, the constrained working space restricts the application of professional instruments, leading to certain parameters being inaccessible for measurement. Secondly, this indirect calibration method may introduce additional calculation errors during the process. Hence, an online, direct optimization method for the decoupling matrix is required. A significant amount of literature has been presented to optimize the parameters of the decoupling matrix. The most essential method to improve decoupling is to obtain the inverse matrix of the steady gain of the plant by means of identification. From the perspective of direct reduction of cross-talk, a iterative optimization method was proposed in [8] to optimize the parameters of decoupling matrix. The elements comprising the SDC can be optimized by making use of step response experiments[9] with actual steady-state gains between the inputs and outputs.

However, there is limited literature addressing the impact of motor dynamic inconsistencies on decoupling accuracy. In reality, manufacturing errors and inherent motor characteristics contribute to changes in certain electrical constants, such as the force constant. The force constant, a pivotal parameter of the motor, elucidates the relationship between coil current and electromagnetic force. In this scenario, while the system is initially completely decoupled in the original position, the dynamic operation of the system introduces changes in characteristics, leading to incomplete decoupling. This phenomenon subsequently impacts the performance of the system in motion.

This paper presents a novel direct data-based dynamic decoupling method, inspired by the virtual reference feedback tuning (VRFT) method[10]. This method is proposed as a solution to address the dynamic differences between motors and the practical errors in mechanical parameters. The key contributions of this work, in comparison to existing literature, can be succinctly outlined as follows:

- (1) The proposed method is a non-iterative data-driven approach that requires minimal model information. It is noteworthy that the necessary measurement data can be obtained through a single experiment.
- (2) The proposed method is a dynamic decoupling approach that considers the inconsistency between actuators.

2 Problem formulation

2.1 Description of the Control System

Consider a precision motion stage which can implement three DOFs motion in vertical direction as depicted in Fig. 1. The simplified structure diagram is shown in Fig. 2, and it can be observed that the precision motion stage is driven by three voice coil motors labeled as m_1 , m_2 and m_3 respectively. The voice coil motor can provide precise and linear motion with high acceleration and high force-to-weight ratio which make it ideal for applications that requires fast and accurate positioning. The measurement system of the stage consists of three linear encoders located at the position near to the motors, for simplicity, suppose that the measured data of the encoders is equal to the displacement of the motors. Furthermore, suppose the controlled point is the center of the

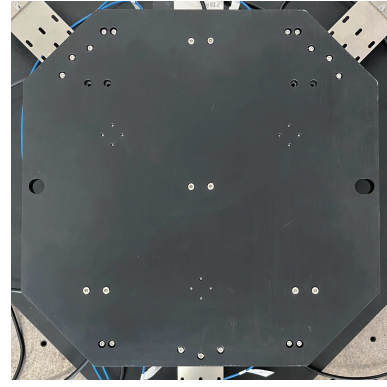


Fig. 1: The picture of the motion stage.

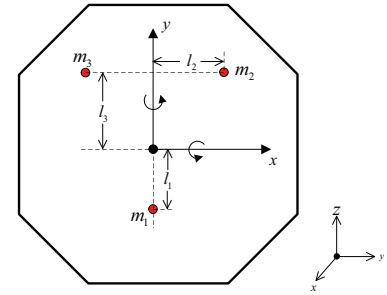


Fig. 2: The diagram of the simplified structure of the motion stage.

stage which refers to performance point and the reference coordinate is C_{xyz} represented in Fig. 2 in which the displacements of the performance point can be obtained directly as y_z , y_{θ_x} and y_{θ_y} respectively. Then the control objective is to achieve three degree-of-freedom trajectory tracking of the performance point in coordinate C_{xyz} .

The configuration of the decoupling control for the motion stage can be depicted as Fig. 3, where $P(s) \in R(s)^{3 \times 3}$ represents the dynamic model with motor driving force as input and measurement data of the linear encoders as output. $T_m \in R^{3 \times 3}$ is the transposition matrix to calculate the three degree-of-freedom of the performance point based on the measurement data of the linear encoders. $G_m \in R^{3 \times 3}$ represents the relationship between the motor instruction and the actual driving force which is subject to current loop and magnetic of the motor. In general, G_m can be treated as a constant matrix. If the mechanical stiffness is sufficiently high and the dynamic characteristics of the motors remain consistent, the static decoupling matrix $K \in R^{3 \times 3}$ can be employed to decouple the system. Subsequently, designing a diagonal feedback controller $C(s) \in R(s)^{3 \times 3}$ becomes relatively straightforward for the decoupled system, which inherently exhibits a diagonal structure. The signal $r(t) \in R^3$ and $e(t) \in R^3$ are high-order polynomial tra-

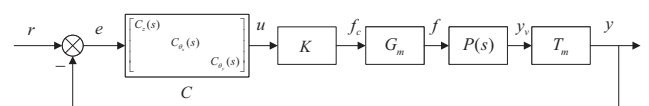


Fig. 3: The decoupling control configuration of the motion stage with multiple DOFs.

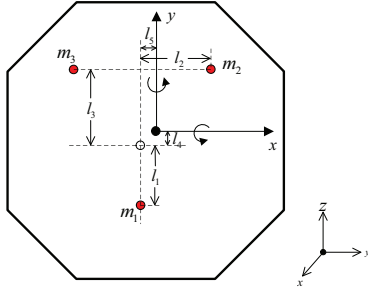


Fig. 4: The actual structure of the short stroke stage with mechanical errors.

jectory and tracking error respectively. $u(t) \in R^3$ denotes the control signal for the decoupled system, calculated by $C(s)$. $f_c(t) \in R^3$ and $f(t) \in R^3$ represent, respectively, the control command of the motors and the actual driving force exerted by the motors. $y_v(t) \in R^3$ is the measurement data of the linear encoders, and $y_t \in R^3$ is the displacement of the performance point in the C_{xyz} coordinate which is determined based on $y_v(t)$. Based on the definition above, the dynamic model of the control system can be described as

$$\begin{cases} y(t) = T_m y_v(t) \\ y_v(t) = P(s) f(t) \\ f(t) = G_m f_c(t) \\ f_c(t) = K u(t) \\ u(t) = C(s) e(t) \\ e(t) = r(t) - y(t) \end{cases} \quad (1)$$

Under the assumption that the motion stage can be treated as a rigid body and with consistent dynamic characteristics of the motors, the static decoupling matrix K can be determined as follows:

$$K = G_m^{-1} \begin{bmatrix} \frac{l_3}{l_1+l_3} & -\frac{1}{l_1+l_3} & 0 \\ \frac{1}{2(l_1+l_3)} & \frac{1}{2(l_1+l_3)} & -\frac{1}{2l_2} \\ \frac{l_1}{2(l_1+l_3)} & \frac{1}{2(l_1+l_3)} & \frac{1}{2l_2} \end{bmatrix}. \quad (2)$$

It is evident that the decoupling matrix K is determined in terms of mechanical parameters and G_m . The resulting decoupled system $T_m P(s) G_m K$ is fully decoupled only if the mechanical parameters are accurate and G_m remains constant throughout the motion. However, during actual processing, assembly, and other manufacturing stages, deviations often occur. Mechanical parameters such as the installation positions of the motors and the actual CoM position may not align precisely with their intended design values as Fig. 4 shows. Additionally, the dynamic characteristics among different motors may be inconsistent, and variations in the characteristics of the same motor can arise during operation. Essentially, due to manufacturing errors, the force constants of different motors may not be uniform. Furthermore, as depicted in Fig. 5, the force constant of the same motor may vary with displacement changes. Based on the observation in Fig. 5, it can be inferred that the force constant can be characterized as a function of the motor's displacement which means G_m should be substituted as

$$G_m^1(y_v) \quad (3)$$

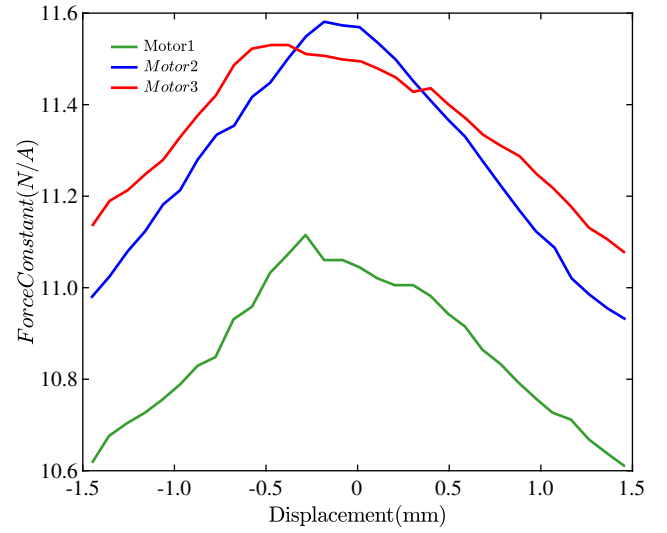


Fig. 5: The diagram of the force constants of the different motors varying with displacement.

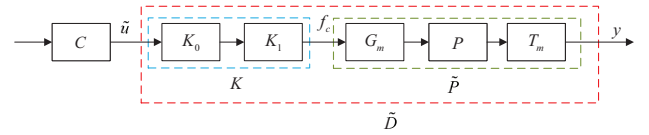


Fig. 6: Block diagram of the proposed method.

At this point, if the decoupling matrix is still selected as (2), there will be non-zero elements off the diagonal in the decoupled system which will lead to the persistence of cross-talk and undermine the intended decoupling effect. To achieve complete decoupling in this case, the ideal decoupling matrix should be modified to K_{ideal} as follows

$$\begin{cases} K_s = \begin{bmatrix} \frac{l_3-l_4}{l_1+l_3} & -\frac{1}{l_1+l_3} & 0 \\ \frac{l_1 l_2 + l_1 l_5 + l_2 l_4 + l_3 l_5}{2(l_1 l_2 + l_2 l_3)} & \frac{1}{2(l_1+l_3)} & -\frac{1}{2l_2} \\ \frac{l_1 l_2 - l_1 l_5 + l_2 l_4 - l_3 l_5}{2(l_1 l_2 + l_2 l_3)} & \frac{1}{2(l_1+l_3)} & \frac{1}{2l_2} \end{bmatrix} \\ K_{ideal} = [G_m^1(y_v)]^{-1} K_s \end{cases} \quad (4)$$

Referring to (4), it is evident that the static decoupling matrix is replaced by a dynamic decoupling controller. This controller comprises a constant matrix for achieving the decoupling of rigid motion and a dynamic component associated with the motor's displacement to compensate for variations in motor characteristics. However, the ideal decoupling controller K_{ideal} hinges on some unknown parameters like l_4 , l_t and $G_m^1(y_v)$. While these parameters can be measured using professional instruments, the process is often cumbersome. Moreover, recalibration may be necessary in the event of operational losses on the motion stage. Consequently, the measurement method employed can significantly impact working efficiency. Thus, there arises a necessity to investigate online parameter optimization methods for dynamic decoupling controllers.

3 Data-driven dynamic decoupling method

In this section, a data-driven dynamic decoupling method is developed to implement the on-line estimation of unknown parameters of dynamic decoupling controller K_{ideal}

to eliminate the cross-talk. From Fig. 5, the force constant of the motor can be approximated as a polynomial function in term of $y_v(t)$ which can be written as

$$G_m^1(y_v(t)) = \begin{bmatrix} g_1(y_{v,1}(t)) & 0 & 0 \\ 0 & g_2(y_{v,2}(t)) & 0 \\ 0 & 0 & g_3(y_{v,3}(t)) \end{bmatrix}. \quad (5)$$

where $y_{v,i}$ is the i -th element of y_v and $g_i(y_{v,i}(t))$ for $i = 1, 2, 3$ is polynomial form with order of m as follows

$$g_i(y_{v,i}(t)) = \alpha_0^i + \alpha_1^i y_{v,i}(t) + \cdots + \alpha_m^i y_{v,i}^m(t) \quad (6)$$

where $\alpha_j^i \in R$ for $j = 0, 1, \dots, m$ and $y_{v,i}^j$ represents the j -th power of $y_{v,i}$. Without loss of generality, $y_{v,i}(t)$ can be neglected, and (6) can be written as g_i . Then the dynamic decoupling controller can be parameterized as

$$K_{ideal} = \begin{bmatrix} \frac{1}{g_1} & 0 & 0 \\ 0 & \frac{1}{g_2} & 0 \\ 0 & 0 & \frac{1}{g_3} \end{bmatrix} \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}. \quad (7)$$

$$= K_1 K_0$$

where $k_{ij} \in R$ for $i, j = 1, 2, 3$ represents the element of K_s . For simplicity, the static part and the dynamic part of the K_{ideal} are denoted as K_0 and K_1 respectively.

At this point, the main problem is to obtain the value of the unknown parameter in (7) including k_{ij} in the static part and coefficients of g_i in the dynamic part. Inspired by virtual reference feedback tuning (VRFT) method which is proposed to auto-tune the feedback controller for SISO systems, a data-driven method is proposed to determine the unknown parameters as aforementioned by using measurement data solely. The configuration of the proposed method is depicted as Fig. 6. As shown in Fig. 6, a reference model is needed which is of diagonal form, and the essential idea is to adjust the parameters of K_{ideal} to make the decoupled system as close to $\tilde{D}(s)$ as possible.

In other words, if the output of $\tilde{D}(s)$ is equal to the output of $\tilde{P}(s)K_{ideal}$ with the same conditions of reference input and feedback controller which means $\tilde{D}(s) = \tilde{P}(s)K_{ideal}$ can be achieved. Suppose that the output of the reference model is equal to $y(t)$, the output of the feedback controller in reference model can be determined as

$$y(t) = \tilde{D}(s)\tilde{u}(s). \quad (8)$$

From (8), if the output of the feedback controller $u(t)$ in actual system can be equal to $\tilde{u}(t)$ by adjusting the parameters of K_{ideal} , the following equations can be established

$$\begin{aligned} \tilde{D}(s)\tilde{u}(s) &= \tilde{P}(s)K_{ideal}u(s) \\ \tilde{D}(s) &= \tilde{P}(s)K_{ideal} \end{aligned} \quad (9)$$

It is obvious that the decoupled system is diagonal completely if (9) is satisfied. Therefore, the following equivalent objective function can be established, and the optimal estimate of the parameters of K_{ideal} can be obtained by minimizing that

$$\begin{aligned} K_{ideal}^* &= \arg \min_{K_{ideal}} J \\ &= \arg \min_{K_{ideal}} \|f(t) - K_{ideal}\tilde{u}(t)\|_2^2. \end{aligned} \quad (10)$$

Substituting (7) into (10), the objective function J can be derived as

$$\begin{aligned} J &= \|f(t) - K_1 K_0 \tilde{u}(t)\|_2^2 \\ &= \|f(t) - \begin{bmatrix} \frac{1}{g_1} & 0 & 0 \\ 0 & \frac{1}{g_2} & 0 \\ 0 & 0 & \frac{1}{g_3} \end{bmatrix} K_0 \tilde{u}(t)\|_2^2 \end{aligned} \quad (11)$$

where g_i is in form of (6) for $i = 1, 2, 3$. Define K_0^i represents the i -th row of K_0 , then (11) can be written as

$$J = \sum_{i=1}^3 \|f_i(t) - \frac{1}{g_i} K_0^i \tilde{u}(t)\|_2^2. \quad (12)$$

where $f_i(t) \in R$ is the i -th element of $f(t)$. Then the minimum of J can be determined by minimizing three independent objective function which can be given as

$$J_i = \|f_i(t) - \frac{1}{g_i} K_0^i \tilde{u}(t)\|_2^2 \quad (13)$$

for $i = 1, 2, 3$.

Substituting (6) into (13), the following equation can be derived

$$\begin{aligned} J_i &= \left\| \frac{1}{g_i} (g_i f_i(t) - K_0^i \tilde{u}) \right\|_2^2 \\ &= \left\| \frac{1}{g_i} ((\alpha_0^i + \alpha_1^i y_{v,i}(t) + \cdots + \alpha_m^i y_{v,i}^m(t)) f_i(t) - K_0^i \tilde{u}) \right\|_2^2 \end{aligned} \quad (14)$$

Because of $\frac{1}{g_i} \neq 0$, $\frac{1}{g_i}$ can be neglected, and Define $y_{v,i}^j(t) f_i(t) = L_j^i(t)$ for $j = 1, \dots, m$. Then the objective function can be simplified as

$$J_i = \|\alpha_0^i L_0^i(t) + \alpha_1^i L_1^i(t) + \cdots + \alpha_m^i L_m^i(t) - K_0^i \tilde{u}(t)\|_2^2 \quad (15)$$

Furthermore, due to $\alpha_0^i \neq 0$, the objective function can be written further as

$$J_i = \|L_0^i(t) + \frac{\alpha_1^i}{\alpha_0^i} L_1^i(t) + \cdots + \frac{\alpha_m^i}{\alpha_0^i} L_m^i(t) - \frac{1}{\alpha_0^i} K_0^i \tilde{u}(t)\|_2^2 \quad (16)$$

To solve the optimization problem, (16) should be converted to discrete form

$$J_i = \frac{1}{N} \|L_0^i(k) + \frac{\alpha_1^i}{\alpha_0^i} L_1^i(k) + \cdots + \frac{\alpha_m^i}{\alpha_0^i} L_m^i(k) - \frac{1}{\alpha_0^i} K_0^i \tilde{u}(k)\|_2^2 \quad (17)$$

where $L_j^i(k)$ and $\tilde{u}(k)$ are the k -th sampling value of $L_j^i(t)$ and $\tilde{u}(t)$ respectively and N is the sampling number. For simplicity, define

$$\begin{cases} T_i = [L_0^i(1), L_0^i(2), \dots, L_0^i(N)]^T \in R^N \\ \phi(k) = [L_1^i(k), L_2^i(k), \dots, L_m^i(k), \tilde{u}(k)]^T \in R^{m+3} \\ \Phi = [\phi(1), \phi(2), \dots, \phi(N)]^T \in R^N \\ \theta_i = [-\frac{\alpha_1^i}{\alpha_0^i}, -\frac{\alpha_2^i}{\alpha_0^i}, \dots, -\frac{\alpha_m^i}{\alpha_0^i}, \frac{K_0^i}{\alpha_0^i}]^T \in R^{m+3} \end{cases}. \quad (18)$$

Substituting (18) into (17), the objective function can be rewritten as

$$J_i = \frac{1}{N} \|T_i - \Phi \theta_i\|_2^2 \quad (19)$$

Then the estimate of θ_i can be obtained by using the least square method as follows

$$\hat{\theta}_i = (\frac{1}{N} \Phi^T \Phi)^{-1} \frac{1}{N} \Phi^T T_i. \quad (20)$$

where the estimate of α_j^i and K_0^i can be obtained from $\hat{\theta}_i$.

Table 1: Mechanical parameters used in the simulation.

Parameters	l_1	l_2	l_3	l_4	l_5
Value(m)	0.1	0.1	0.1	0.01	0.01

4 Simulations Results

In this section, the effectiveness of the proposed method is validated through comparing the nominal decoupling method and the ideal static decoupling method which can decouple the system in case of no difference among motors, and these methods can be represented as follows:

- (1) M_0 : The nominal static decoupling control method with decoupling matrix of K_0 .
- (2) M_1 : The ideal static decoupling control method with decoupling matrix of K_1 .
- (3) M_2 : Proposed data-based dynamic decoupling control method with dynamic decoupling controller of K_2

4.1 Simulation Setup

A simulation scheme is responsible for illustrating the effectiveness of the proposed method as Fig. 2 shows. The value of mechanical parameters are shown in Table. 1. Suppose that the mass of the SS stage is $m = 4.5862kg$ and the moment of inertia of θ_x and θ_y at the CoM are $I_x = 0.0484kg \cdot m$ and $I_y = 0.0414kg \cdot m$ respectively. Under the rigid body hypothesis, the transfer function model between the driving forces of motors and the three DOFs displacement of the CoM can be given as

$$P(s) = \begin{bmatrix} \frac{0.2180}{s^2} & \frac{0.2180}{s^2} & \frac{0.2180}{s^2} \\ -\frac{2.2727}{s^2} & \frac{1.8595}{s^2} & \frac{1.8595}{s^2} \\ \frac{0.2415}{s^2} & -\frac{2.1739}{s^2} & \frac{2.6570}{s^2} \end{bmatrix} \quad (21)$$

From (2), the nominal static decoupling matrix K_0 can be determined as

$$P(s) = \begin{bmatrix} 0.5 & -5.0 & 0 \\ 0.25 & 2.5 & -5.0 \\ 0.25 & 2.5 & 5.0 \end{bmatrix} \quad (22)$$

Then the desired decoupled model can be given as

$$D(s) = \begin{bmatrix} \frac{0.2180}{s^2} & 0 & 0 \\ 0 & \frac{20.6612}{s^2} & 0 \\ 0 & 0 & \frac{24.1546}{s^2} \end{bmatrix}. \quad (23)$$

Successively, the diagonal PID feedback controller $C(s)$ can be easily tuned in term of the diagonal elements of $D(s)$ and since the nominal decoupled system is diagonally dominant even if not completely decoupled, the stability of the actual system can be obtained with $C(s)$. Nevertheless, due

to the inaccurate mechanical parameters, there will be serious performance deterioration of performance. To decouple the system completely in this case, the ideal static decoupling matrix K_1 can be determined according to (4) as

$$K_1 = \begin{bmatrix} 0.45 & -5.0 & 0 \\ 0.325 & 2.5 & -5.0 \\ 0.225 & 2.5 & 5.0 \end{bmatrix} \quad (24)$$

Then, the decoupled system with K_1 is diagonal without the dynamic differences between motors. However, due to the errors in manufacturing, the dynamic characteristics of the motors are inevitably different which can result in incomplete decoupling and lead to deteriorating cross-talk. This issue can be solved by the dynamic method proposed in this paper. According to the measuring data, the force constant of three voice coil motors can be approximated as third order polynomials, and the parameters are listed in Table. 2.

Table 2: Parameters of the force constant polynomial used in the simulation.

Parameters (N/m^i)	m_1	m_2	m_3
α_0	11.02	11.49	11.49
α_1	-52.99	-90.72	-100.08
α_2	-2.04×10^5	-2.74×10^5	-1.83×10^5
α_3	2.45×10^7	3.42×10^7	4.01×10^7

In M_2 , the reference $\tilde{D}(s)$ model can be chosen as $D(s)$, and the order used in (6) can be determined as $m = 2$. By conducting excitation experiments on the stable closed-loop system using K_0 , the dynamic decoupling controller K_2 can be determined, relying on the actual displacements of motors.

4.2 Performance assessment

To visually demonstrate the effectiveness of M_2 , validation experiments were conducted using various decoupling controllers. The decoupling performance of different methods was compared in the time domain by assessing the maximum value of cross-talk. Elaborating further, tracking experiments for three DOFs were performed utilizing the same feedback controller across different decoupled systems. In the tracking experiments, the step signals of $100\mu m$, $100\mu rad$ and $100\mu rad$ are tracked by z -DOF, θ_x -DOF and θ_y -DOF respectively, while the reference input of other DOFs are set to 0, and experiment results are shown in Fig. 7, Fig. 8 and Fig.9 respectively. It can be observed that the M_2 exhibited the best performance, and the cross-talk in the decoupled system with K_2 is almost negligible compared to other methods.

5 Conclusion

The precision of the decoupling controller is based on the mechanical parameters and the dynamical characteristics of actuators which is commonly neglected. However, in the field of precision motion control, cross-talk resulting from inaccurate decoupling has emerged as a significant obstacle to achieving high performance. Hence, it is imperative to

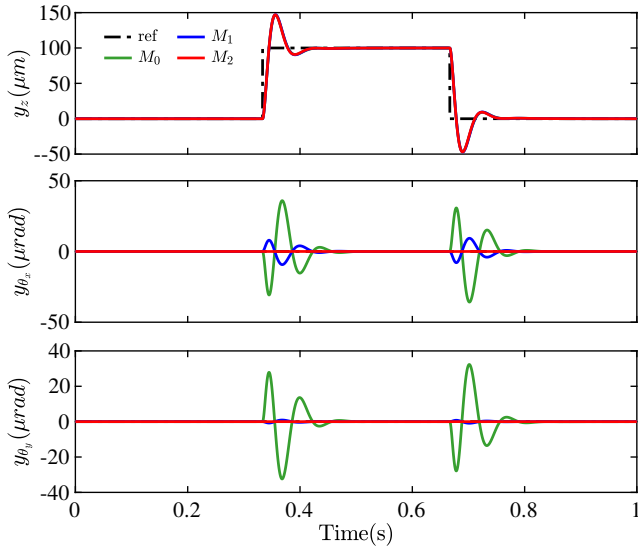


Fig. 7: The output responses of tracking experiment of z -DOF.

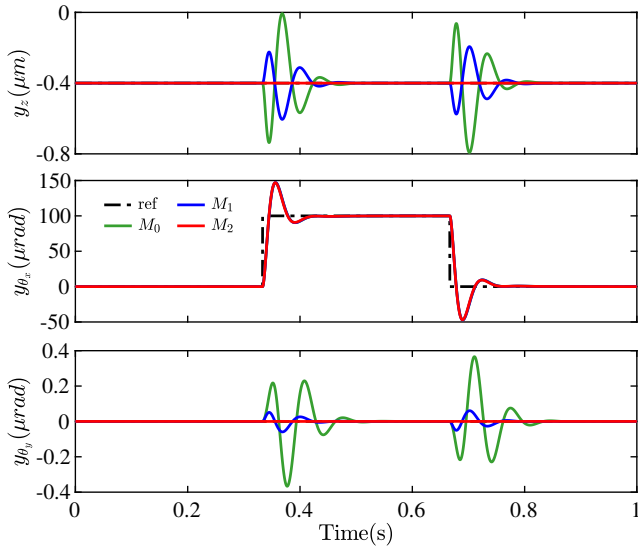


Fig. 8: The output responses of tracking experiment of θ_x -DOF.

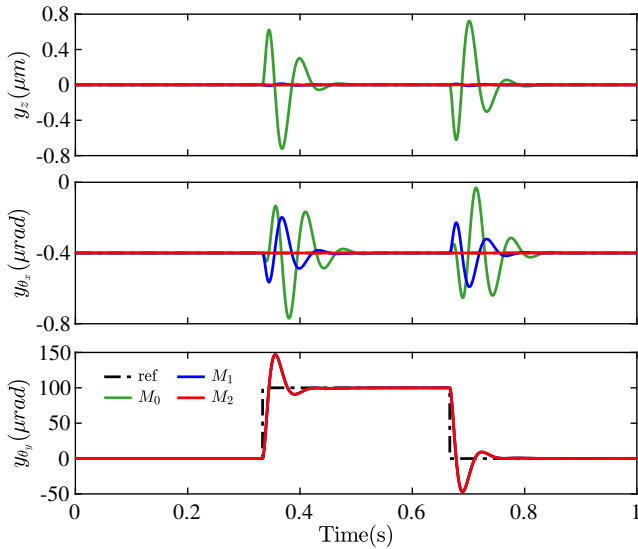


Fig. 9: The output responses of tracking experiment of θ_y -DOF.

consider the inconsistent characteristics of different motors when designing the decoupling controller. This paper analyzes the dynamic characteristics of motors and introduces a dynamic decoupling controller to enhance the precision of decoupling by compensating for differences between different motors. A data-driven method is developed to obtain the parameters of the dynamic decoupling controller, requiring only a single experimental setup. The effectiveness and superiority of the proposed method are demonstrated through simulations on a precision motion stage.

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